

Relax and Recover: Disaster Recovery for UEFI Systems

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Integrating UEFI into Relax-and-Recover

by

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<http://j.mp/rear-uefi>

Backup != Restore/Recovery



Linux and Disaster Recovery

Disaster Recovery Strategies

➔ Disk Imaging

- + Offline -> no open file issues
- + Simple to restore
- ➔ Online -> very problematic
- ➔ No “perfect” open source tool for Linux available

➔ Copy files and store disk layout and boot info

- + Online -> no problem
- + Backup independent of disk layout and sizes
- ➔ More effort required to restore (can be scripted)
- ➔ Maybe consistency problems, but should be solved by backup solution

Advantages of a Linux System

- ➔ All information is stored in files, all files always readable
- ➔ Operation System and applications can be “slim” - 600MB enough for complete standard server
- ➔ Open system – open methods and procedures
- ➔ All steps of an installation can be scripts: Partitioning, file systems, boot loader etc.

Disaster Recovery - Media

- ➔ Most important: External storage!
- ➔ Bootable media: CD/DVD, USB key, LAN, tape ...
- ➔ Media usually combination boot and backup media:
 - ➔ Bootable CD/DVD, USB key with backup data on it
 - ➔ LAN boot (PXE) with backup data via CIFS, NFS ...
 - ➔ Bootable tapes - HP OBDR (CD emulation)
- ➔ Separation between boot media and backup data
 - ➔ Boot the system from a (small) USB key, CD/DVD or LAN
 - ➔ Recover the system with backup software, tar, rsync ...

Disaster Recovery - How It Works

- ➔ Store the disk layout
 - ➔ Partitioning, LVM and RAID configuration
 - ➔ File systems, file system labels ...
 - ➔ Boot loader (GRUB, GRUB2, LILO, ELILO)
- ➔ Store the files (tgz, rsync, through backup software ...)
- ➔ Create bootable rescue media with system configuration (and backup data)
- ➔ Can be done online
 - ➔ No business interruption
 - ➔ 100% compatible with original systems hard- and software

Disaster Recovery - Rescue Media

- ➔ Create “rescue linux” from running system
- ➔ Optimally compatible “tool box”
- ➔ Clone the system environment
 - ➔ Linux kernel and modules
 - ➔ Device driver configuration
 - ➔ Network configuration
 - ➔ Basic system software and tools
- ➔ Operate entirely in RAM (initrd)



Disaster Recovery - In Action

- ➔ Boot system from rescue media
- ➔ Restore disk layout
 - ➔ Create partitions, RAID configuration and LVM
 - ➔ Create file systems (mkfs, mkswap)
 - ➔ Configure file systems (labels, mount points)
- ➔ Restore the backup data
- ➔ Restore the boot loader
- ➔ Reboot
- ➔ Done!

Relax and Recover



Relax and Recover - Rear



- <http://relax-and-recover.org/>
- GPL Software – Developers in Germany and Belgium
- 100% Bash script – no GUI and no dependencies
- Utilize kernel, modules, binaries of host (kernel ≥ 2.6)
- Support any combination of SW/HW RAID, LVM
- Internal backup on CIFS, NFS, rsync ...
- Boot media on CD/DVD, USB storage and LAN (PXE)
- Bootable tapes

- Successor of mkCDrec

Rear - Features

- ➔ Focus on disaster recovery and **not** backup
- ➔ Tight integration with common backup software – delegate file backup to backup infrastructure
- ➔ Simple full backup integrated
- ➔ Complements backup software:
 - ➔ Backup software: Data storage and retrieval
 - ➔ ReaR: Recover system layout and make it work again
 - ➔ ReaR utilizes the backup software to restore the backup data
- ➔ Use the best tool for the job

Rear - Backup Software

- ➔ Supported solutions include:
 - ➔ CommVault Galaxy
 - ➔ IBM Tivoli Storage Manager
 - ➔ Veritas NetBackup
 - ➔ HP Data Protector
 - ➔ Bacula
 - ➔ Duplicity (experimental)
 - ➔ Rsync and other “external” methods
 - ➔ tar.gz archive on NAS share – CIFS, NFS, NCP ...
- ➔ Very transparent integration
- ➔ Quick implementation: 2-3 PT programming
- ➔ Can be easily extended to support other vendors

Rear - Network Integration

- ➔ Disaster recovery as part of network infrastructure
 - ➔ Backup software – file-level backup storage
 - ➔ Rear – system environment
 - ➔ Boot rescue media via PXE – no physical media required
 - ➔ Very scalable – automated installation of entire disaster recovery data center



Rear - Status

- ➔ Stable software
 - ➔ i386 and x86_64 well tested
 - ➔ ia64 and ppc works, but less tested
 - ➔ UEFI in rear > 1.14-git
- ➔ Regular releases (RPM, DEB, TGZ)
- ➔ Major Linux distributions ship Rear:
 - ➔ **SLES11 >= SP1 HA Extension**
 - ➔ OpenSUSE >= 11.2 and Fedora >= 11
- ➔ Community and commercial support available
- ➔ Regular patch submissions from Rear users

ReaR - Development

- ➔ Open Source development model:
 - ➔ Submit patches and feedback – “field testing”
 - ➔ Sponsoring
- ➔ Modular design:
 - ➔ Rear is a framework to plug together many small Bash scripts
 - ➔ Maximize code reusability
 - ➔ Simple development model (vi works fine)
 - ➔ Little to no “interferences” between different areas of code
- ➔ Documentation on project homepage
 - ➔ Hierarchical structure
 - ➔ References to source code



Integration of UEFI into rear (ia64)

- Integrity platform (ia64) UEFI support was added long time ago
- Using the UEFI standard v1 or v2 (**no secure boot**)
- What do we need to integrate?
 - **/boot/efi** : mounted as vfat
 - **/boot/efi/efi/*/elilo.efi** : boot loader (same for different flavors of Linux)
 - `CONSOLE="console=tty1 console=ttyS1"` : mandatory
 - No need to be grubby after recovery as `/boot/ef/*` is all you need
 - Create a bootable CDRROM which is recognized by UEFI

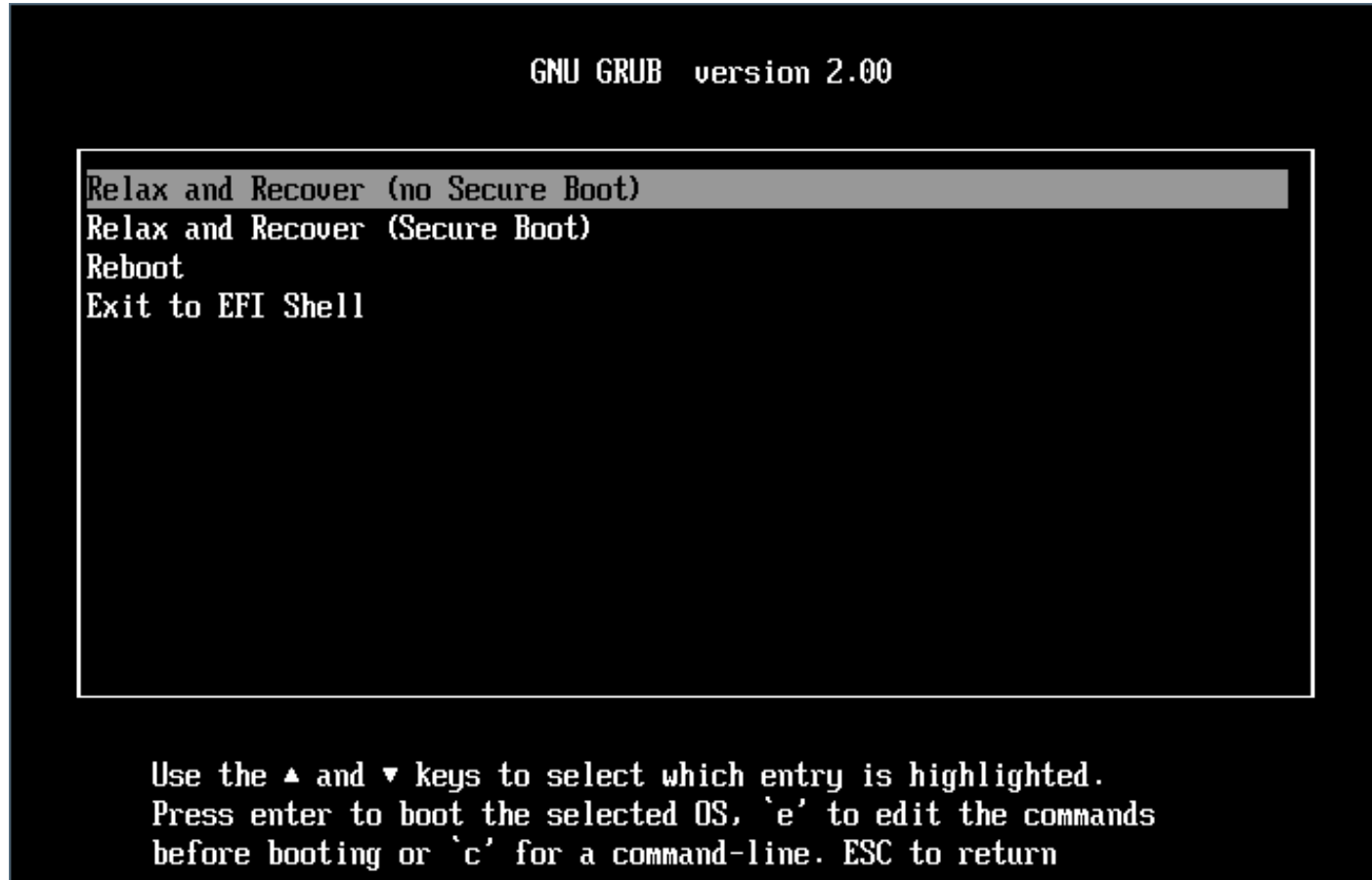
Integration of UEFI into rear (x86_64)

- ➡ What do we need for UEFI support on Linux?
 - ➔ Bootable disk with GPT partition table (parted /dev/disk p)
 - ➔ /boot/efi mount point (vfat)
 - ➔ Linux Kernel Config should contain CONFIG_EFI=y
 - ➔ UEFI Runtime Variables/Services Support - 'efivars' kernel module
 - ➔ Check /sys/firmware/efi/vars/ directory
 - ➔ Efibootmgr to manipulate boot entries, order of booting
 - ➔ Create a bootable UEFI capable ISO image

UEFI / GPT notes

- ➔ To manipulate disk devices with GPT label we need
 - ➔ Be sure this system uses UEFI
 - ➔ Parted (./conf/Linux-i386.conf:parted) 
 - ➔ Gdisk (GPT fdisk utility – not mandatory, but nice to have)
 - ➔ A mounted /boot/efi file system (type vfat)
 - ➔ The efivars kernel module
 - ➔ Efibootmgr utility
 - ➔ Which boot manager is used (grub, elilo, gummiboot, shim,...)
 - ➔ Secure boot used? Recovered system might be unbootable!

Hybrid ISO



EFI Support in rear

```
~/src/rear$ find . -iname \*efi\*
```

```
usr/share/rear/conf/templates/EFI_readme.txt
usr/share/rear/prep/default/31_include_uefi_tools.sh
usr/share/rear/prep/ISO/Linux-ia64/34_define_console_ia64.sh
usr/share/rear/prep/ISO/Linux-ia64/33_find_elilo_efi.sh
usr/share/rear/prep/OBDR/Linux-ia64/34_define_console_ia64.sh
usr/share/rear/prep/OBDR/Linux-ia64/33_find_elilo_efi.sh
usr/share/rear/output/default/15_save_copy_of_prefix_dir.sh
usr/share/rear/output/default/20_make_prefix_dir.sh
usr/share/rear/output/ISO/Linux-i386/20_mount_efibootimg.sh
usr/share/rear/output/ISO/Linux-i386/70_umount_efibootimg.sh
usr/share/rear/output/ISO/Linux-i386/25_populate_efibootimg.sh
usr/share/rear/output/ISO/Linux-ia64/40_create_local_efi_dir.sh
usr/share/rear/output/RSYNC/default/20_make_prefix_dir.sh
usr/share/rear/output/OBDR/Linux-ia64/40_create_local_efi_dir.sh
usr/share/rear/lib/uefi-functions.sh
usr/share/rear/finalize/Linux-i386/23_run_efibootmgr.sh
usr/share/rear/backup/NETFS/default/15_save_copy_of_prefix_dir.sh
usr/share/rear/backup/NETFS/default/20_make_prefix_dir.sh
usr/share/rear/rescue/default/85_save_sysfs_uefi_vars.sh
```

UEFI Status

- ➔ Currently in HEAD, part of next release
- ➔ Tested on Fedora 18, Ubuntu 12.10
- ➔ OpenSuse 12.2 (and 12.3) failed to generate a bootable UEFI ISO
https://bugzilla.novell.com/show_bug.cgi?id=811636
- ➔ Secure Boot
 - ➔ only working on same system (Key Exchange Keys - KEKs)
 - ➔ P2P, P2V is not possible due to the KEKs
- ➔ Secure Boot disabled works out of the box

Start from the sources

- ➔ `$ git clone git@github.com:rear/rear.git`
- ➔ `# yum|zypper install rpm-build lsb mingetty`
- ➔ `$ make rpm`
- ➔ `$ sudo rpm -ivh rear-1.14-1.git201303211657.noarch.rpm`
- ➔ `$ sudo -i`
- ➔ Rear is at your service:
 - ➔ `/etc/rear/local.conf`
 - ➔ `/usr/share/rear/*`
- ➔ Edit `/etc/rear/local.conf`:
 - ➔ `BACKUP=NETFS`
 - ➔ `OUTPUT=ISO`

Writing your own rear scripts

- ➔ Good to know – everything is a script, even config files
- ➔ Does rear has an API? Yes, check out our functions:
`grep '()' /usr/share/rear/lib/*functions.sh`
- ➔ Rear works with workflows – see other presentations on the basics
- ➔ Where to drop your script? Use 'rear -s mkbackup' to see all existing scripts and order of execution



Relax-and-Recover
rear

 Transcontinental

 <http://relax-and-recover.org/>

 Joined on Jan 29, 2012

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Thank you very much! Please contact me for further questions and discussions.



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